Dear Quillstone Payments AI Team,

I am applying for the Machine Learning Engineer – Customer Support AI role. At Thistlebox, I build and operationalise machine-learning workflows that support business decisions, and I am keen to bring that experience to Quillstone's work on helping merchants get support quickly and with less friction.

I developed a scheduled LightGBM forecasting workflow in Vertex AI Workbench, producing six-week-ahead forecasts in BigQuery for operational use in Looker. It improved WAPE from 23% to 14% and contributed to an estimated 9% reduction in food waste. I also built an XGBoost churn propensity model with feature engineering and SHAP analysis, with weekly batch scores in BigQuery integrated into dbt workflows for retention targeting. In a holdout, churn fell by 2.1 percentage points in the treated group.

Alongside modelling, I built a Streamlit application for planning and analysing sequential A/B tests, writing the application and its Dockerfile; the data platform team handled deployment. I have also started writing pytest unit tests for forecasting feature code, with coverage now at approximately 60%. My daily work with Python, SQL and BigQuery, together with my use of Vertex AI Workbench and dbt, has given me practical experience building data and scoring workflows. My MSc in Statistics provided a foundation in experimental design and modelling uncertainty.

Quillstone's focus on customer-support automation and its emphasis on ownership, teamwork and learning from mistakes are appealing to me. I would welcome the opportunity to contribute my experience in forecasting, churn modelling and experimentation while continuing to develop as a machine-learning engineer.

Thank you for considering my application.

Kind regards,
Tobias Mwangi-Hartley